

Passive DS-TWR Deadline Pipeline and Rolling Solver Field Validation

UWB Module ESP-IDF field test

28 July 2026

Abstract

This report validates the non-blocking DW3000 deadline pipeline and the low-latency rolling position solver added to the custom receive-only Passive DS-TWR protocol. In a static surveyed geometry, deadline scheduling increased independent position throughput from approximately 63 to 84 updates/s without an observed accuracy regression. Rolling solving preserved the independent frame rate while increasing total display updates to 169/s. The deadline pipeline also reduced combined RESP/FINAL timeout counts by approximately 73% and removed the large anchor-range bursts present in the legacy millisecond/blocking scheduler during this test.

1 Objective

The previous Passive DS-TWR speed-limit study identified the radio transaction and millisecond receive-loop scheduling path as the principal bottleneck. It recommended four changes:

1. instrument POLL TX, delayed RESP TX, FINAL TX/RX, CIA readout, and RX re-arm;
2. replace millisecond receive-loop scheduling with DW3000 delayed-TX and deadline-driven state transitions;
3. retain Robust Rotating at 1.0/1.0 ms as the accuracy control;
4. evaluate rolling solves as a distinct low-latency display mode while reporting independent frames/s separately.

Firmware 7226cca implements those changes. The purpose of this field test is to determine whether they improve speed without sacrificing static accuracy or destabilising the radio schedule.

2 Experimental method

Five modules were flashed once in one simultaneous OTA operation. Four anchors formed the known 3 m × 3 m reference square: A2 = (0, 0), A3 = (0, 3), A4 = (3, 0), and A5 = (3, 3) m. Tag 1 was stationary at the surveyed centre, (1.5, 1.5) m. Dynamic anchor self-localisation remained active, matching the intended operational system rather than forcing fixed dashboard coordinates.

Every test used Robust Rotating with 1.0 ms RESP and 1.0 ms FINAL timing. Four 60 s blocks, each preceded by a 10 s warm-up, formed an A–B–C–A sequence:

A1 Legacy receive loop, one solve per complete frame;

B deadline pipeline, one solve per complete frame;

C deadline pipeline, rolling solves capped at 100 Hz;

A2 return to the Legacy control.

Positions were captured from the dashboard's lossless event stream. Raw solver and constant-velocity EKF outputs were retained separately. Accuracy for the rolling block is calculated only from its complete independent-frame solutions; correlated intermediate rolling results are not counted as additional independent accuracy samples.

3 Throughput

Mode	independent frames/s	rolling updates/s	total solver updates/s	independent samples
Legacy + Frame A1	62.54	–	62.54	3,753
Deadline + Frame	84.12	–	84.12	5,053
Deadline + Rolling	84.38	84.76	169.14	5,066
Legacy + Frame A2	63.24	–	63.24	3,799

Table 1: Measured position throughput over each 60 s capture.

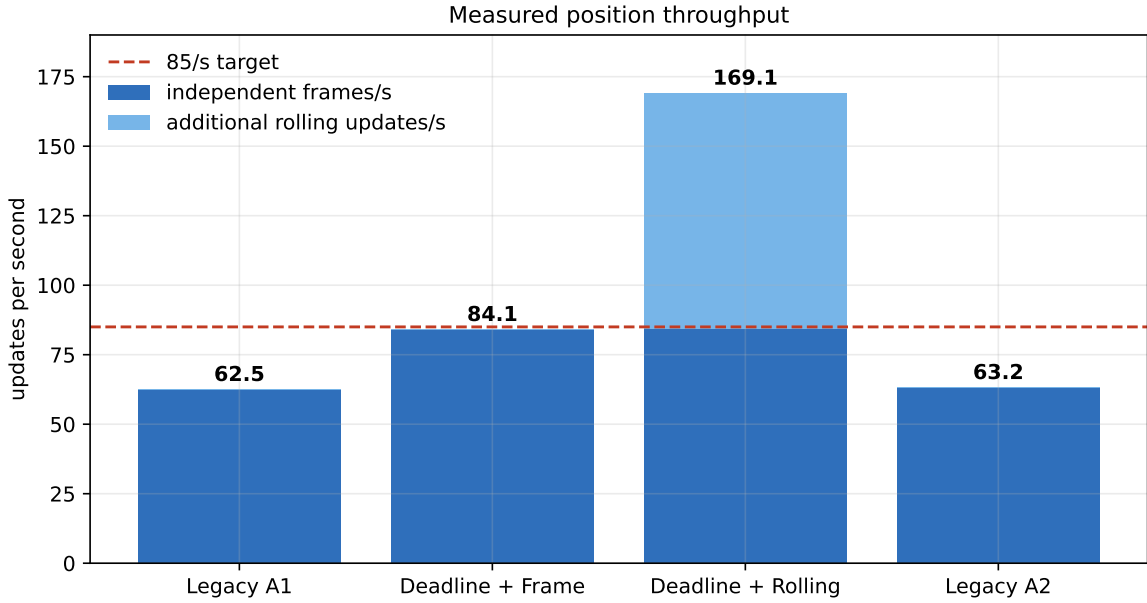


Figure 1: Independent frame rate and additional correlated rolling updates. The rolling solver approximately doubles display refresh while leaving radio frame throughput unchanged.

Deadline scheduling increased independent frame throughput by 34.5% over A1. The A2 return control reproduced the Legacy rate, which supports attributing the gain to the new scheduling pipeline rather than a transient change in network or dashboard performance. Enabling the rolling solver changed the independent rate from 84.12 to 84.38 frames/s; consequently, there is no measured evidence that the added solver work slows the ranging path.

4 Static position accuracy

Mode	EKF error [cm]		raw error [cm]		EKF bias [cm]	
	RMSE	P95	RMSE	P95	x	y
Legacy + Frame A1	3.59	5.51	3.87	6.17	+0.30	-3.12
Deadline + Frame	3.23	5.09	3.59	5.92	+1.04	-2.47
Deadline + Rolling	3.01	4.72	3.28	5.32	-0.05	-2.43
Legacy + Frame A2	5.12	7.29	5.35	8.04	+1.62	-4.50

Table 2: Position errors relative to the surveyed centre. Deadline + Rolling uses only independent-frame solutions in this accuracy table.

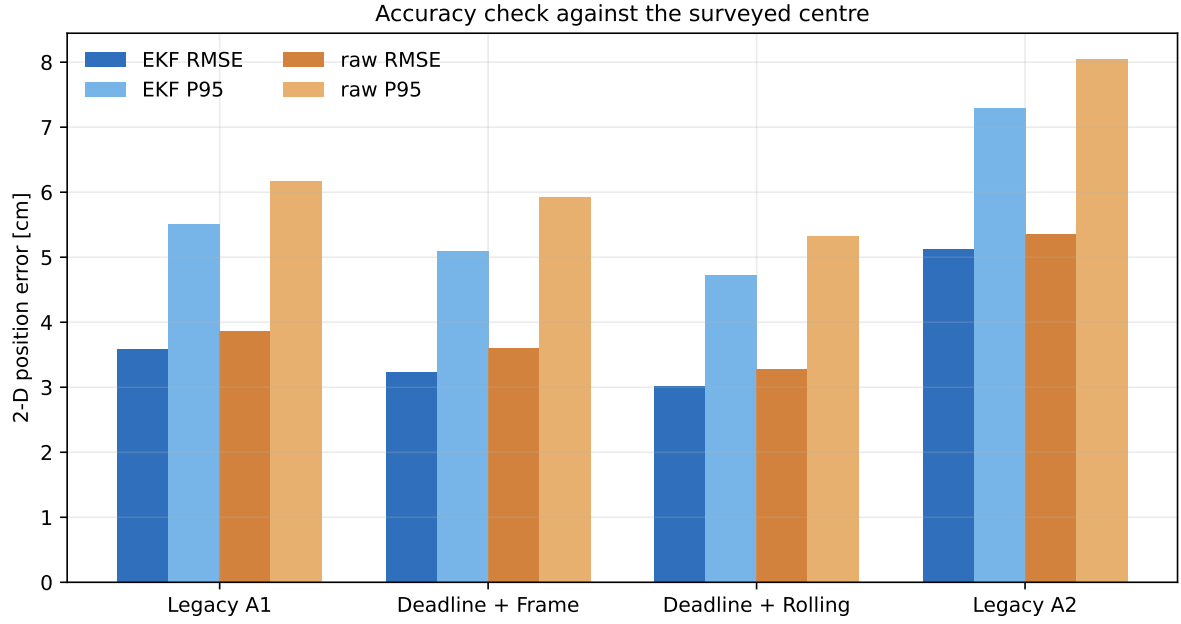


Figure 2: Filtered and raw accuracy. The raw results confirm that the accepted deadline result is not solely an EKF artefact.

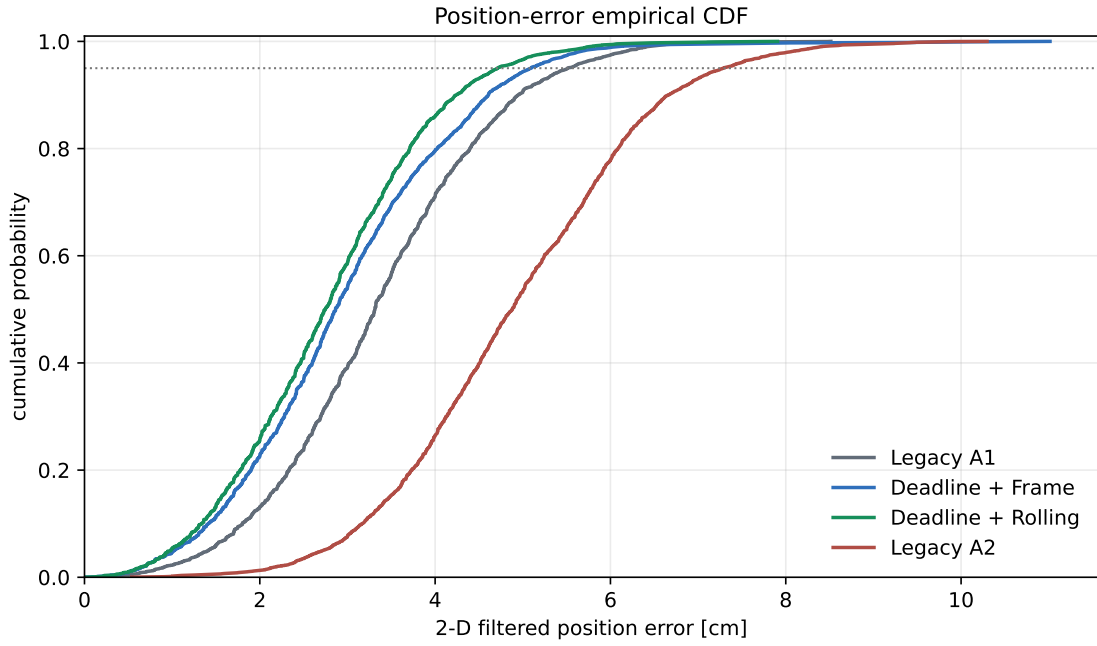


Figure 3: Empirical CDF of filtered 2-D position error.

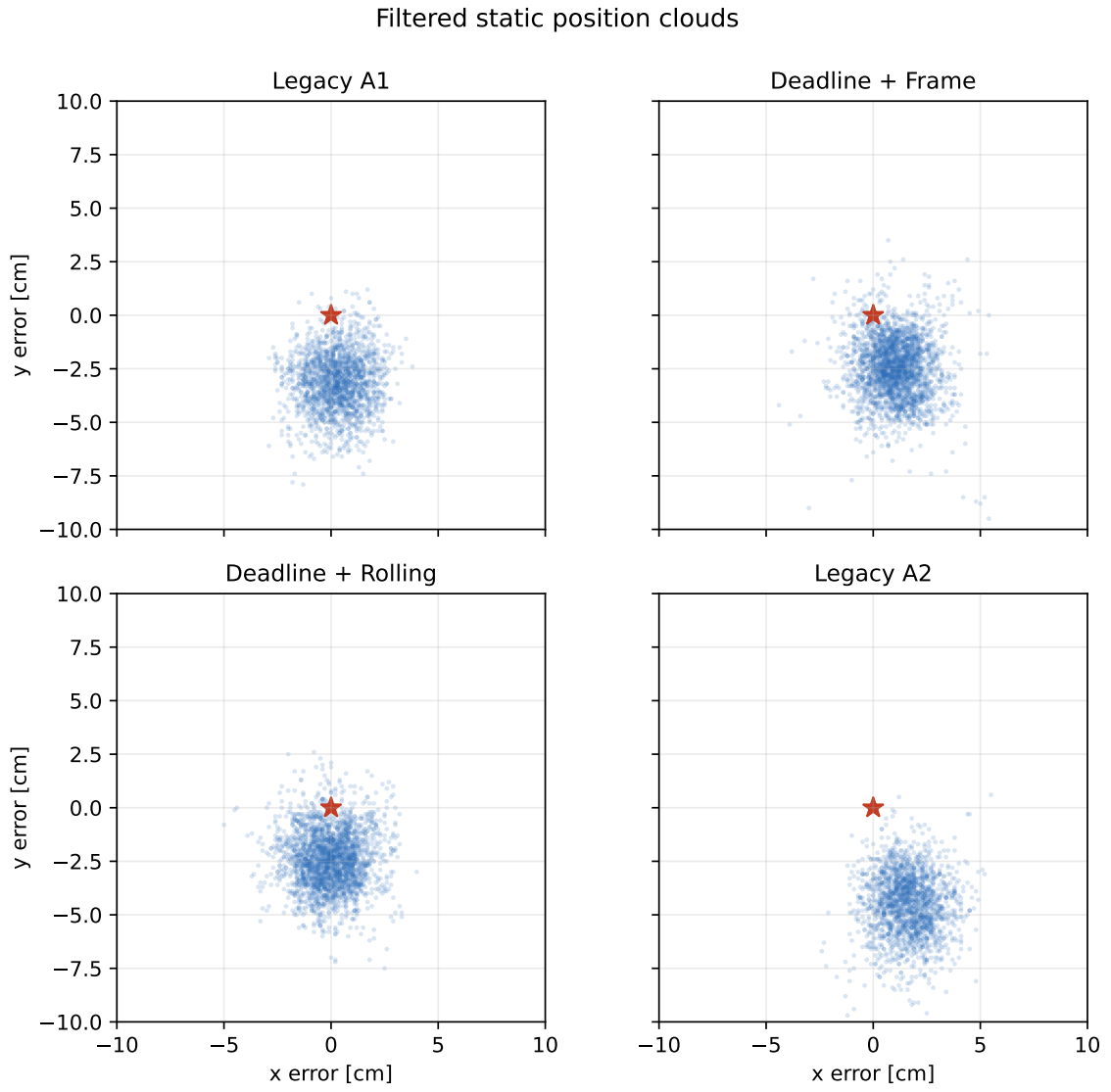


Figure 4: Filtered position clouds relative to the surveyed centre.

4.1 Confidence-band acceptance

A moving-block bootstrap with 3,000 iterations was used to retain temporal correlation. The acceptance rule requires both the candidate point RMSE and P95 to remain no greater than the upper 95% confidence bound of the same-geometry control.

Comparison	metric	control [cm]	control upper 95% bound [cm]	candidate [cm]	result
Deadline Frame vs Legacy A1	RMSE	3.59	3.71	3.23	accepted
	P95	5.51	5.76	5.09	accepted
Rolling frames vs Deadline Frame	RMSE	3.23	3.36	3.01	accepted
	P95	5.09	5.30	4.72	accepted

Table 3: Predefined accuracy gates.

Both candidate configurations pass. These data demonstrate no observed accuracy regression; they do not yet prove that deadline scheduling improves accuracy in all conditions. The captures were sequential and the live geometry estimator continued to evolve.

5 Radio-pipeline diagnostics

Mode	complete	RESP TO	FINAL TO	collision	overrun	invalid	re-arm fail
Legacy + Frame A1	23,791	1,914	619	0	0	0	0
Deadline + Frame	32,128	408	277	672	39	0	0
Deadline + Rolling	32,585	305	275	620	41	0	0

Table 4: Counters accumulated across all modules during each 60 s interval. TO denotes timeout.

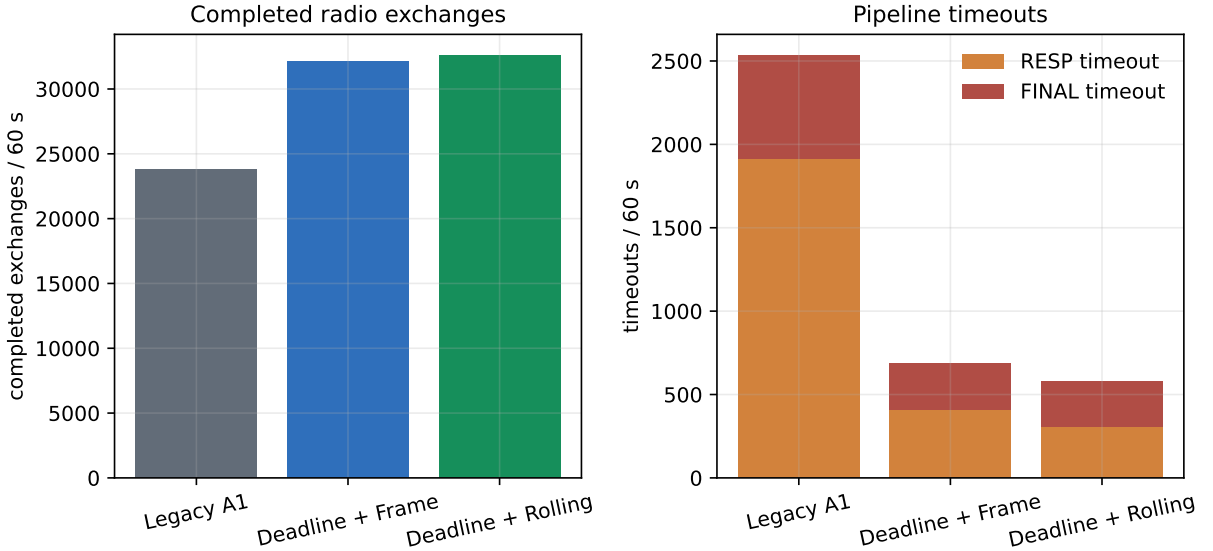


Figure 5: Completed exchanges and timeout counts.

Compared with Legacy A1, Deadline + Frame completed approximately 35% more exchanges and reduced combined RESP/FINAL timeout counts from 2,533 to 685, a reduction of approximately 73%. No invalid frame or RX re-arm failure was reported. CIA readout averaged approximately 25–42 μ s across modules, while RX re-arm averaged 38–46 μ s.

The deadline schedule exposes state collisions and rare schedule overruns that were not explicit counters in the Legacy path. They remain useful optimisation targets, but did not prevent the

measured throughput gain. Timer alarm counts are normal deadline wake-ups and are not failure events.

6 Anchor-geometry stability

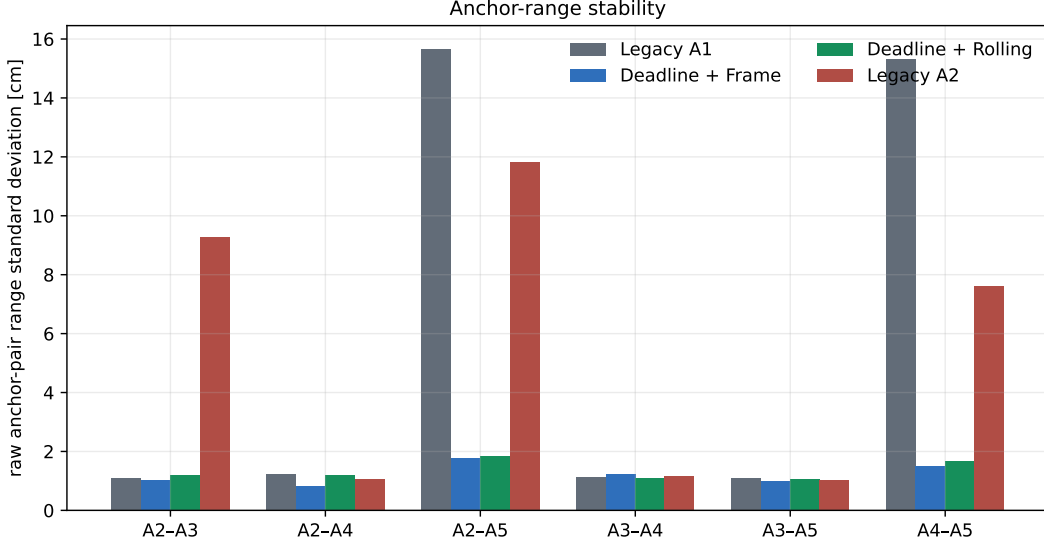


Figure 6: Standard deviation of raw anchor-pair DS-TWR ranges.

Deadline + Frame kept every anchor-pair standard deviation at or below 1.78 cm. Both Legacy captures contained intermittent bursts. In A2, standard deviation reached 9.27 cm on A2-A3, 11.81 cm on A2-A5, and 7.62 cm on A4-A5. These bursts explain the degraded A2 position cloud and match the short-lived geometry deformations previously observed in the dashboard.

Mean anchor-pair biases of several centimetres remain. They are a separate antenna/pair-calibration issue and should not be attributed to deadline scheduling. The present experiment evaluates relative scheduling behaviour, not a new absolute calibration.

7 Conclusions

1. The DW3000 deadline pipeline reaches approximately **84 independent positions/s**, essentially the previously observed FlexTDOA target rate, while keeping the native 1.0/1.0 ms radio timing.
2. Both filtered and raw accuracy gates pass. No accuracy concession was required to obtain the throughput gain.
3. Rolling solving delivers approximately **169 display updates/s** without reducing independent frame throughput. Its additional results are correlated and must continue to be reported separately.
4. Deadline scheduling materially reduces timeout counts and anchor-range bursts. This should also improve the stability of live anchor self-localisation.
5. Remaining work should focus on the recorded state collisions, rare schedule overruns, and static pair/antenna bias calibration. A longer randomised or interleaved field campaign is required before claiming a general accuracy improvement.

Operational state and reproducibility. All modules were left online in Deadline + Rolling at 100 Hz with Robust Rotating 1.0/1.0 ms; the dashboard remained active. The report directory contains all captures, metadata, timing logs, bootstrap outputs, metrics, and the reproducible report generator in `tools/`.